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A Devonian palaeomagnetic pole for Armorica

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Summary. The palaeomagnetic study of Siluro–Devonian redbeds of the San Pedro formation, at Cabrillanes, Asturias, Spain, reveals a high temperature and pre-tectonic magnetization determined by intersection of great circles as well as conventional averaging (Fisher). This magnetization is carried by the haematitic pigment and is therefore of chemical and post-depositional origin. The formation mean direction ($D = 113°$, $I = +34°$, $\alpha_{95} = 10°$) is interpreted as representing a record of a Devonian palaeofield. The corresponding pole after correction for the opening of the Bay of Biscay and the tectonic closure of the Ibero–Armorican arc is in good agreement with contemporaneous European poles, confirming that the Armorica plate had already joined the Laurentia–Baltica assembly at that time.

Introduction

In order to explain the first palaeomagnetic data obtained within the European and North American late Palaeozoic orogenic belts, Van der Voo (1982) has introduced a new plate, called Armorica, allowed to move independently between the three major plates involved in the geodynamic processes of the assembly of Pangaea (Laurentia, Baltica and Gondwana). The geographic extent and the Palaeozoic evolution of this element have yet to be defined more precisely, especially for the Ordovician–Devonian time span where only a few data are available. We report here a palaeomagnetic result from redbeds (San Pedro Formation) having a stratigraphical age spanning the Siluro–Devonian boundary and sampled in the Asturian arc in Spain. For a synthesis of the data available at present for the Armorica plate, we refer the reader to Perroud, Bonhommet & Van der Voo (1983). The present study has been conducted to test the relationships between structural parameters and palaeomagnetic directions in a plunging fold as well as to determine a palaeopole. The magnetizations reported here present a positive fold test and satisfy all the quality criteria usually required for palaeomagnetic synthesis, and are therefore considered to be usable for geodynamic considerations.

Geological setting

The sampling area (Fig. 1) (locality of Cabrillanes, south limb of the Asturian arc, northern Spain) belongs to the tectonic setting of the Ibero–Armorican arc, a large-scale geological

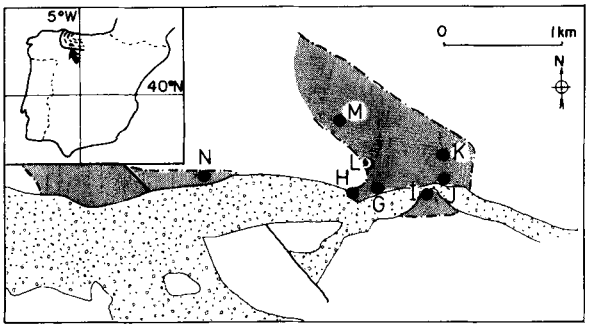


Figure 1. Schematic map of the sampling area, showing the sites (black dots) distributed all around the cartographic nose of a plunging fold, within the San Pedro Formation (darked area). The fold is partially overlain by recent alluvian (dotted areas). Inset is a map of Spain showing the Asturian arc and the locality of Cabrillanes (arrow).

feature connecting the Iberian and Armorican Massifs across the Bay of Biscay. The evidence which has identified that structure is palaeontologic as well as stratigraphic, petrographic and tectonic (Matte 1974; Cogné 1971; Paris & Robardet 1977). Previous palaeomagnetic studies (Ries, Richardson & Shackleton 1980; Perroud & Bonhommet 1981) have shown that this arc had a partially secondary (tectonic) origin representing 80° of the total 150° of arc (Perroud & Bonhommet 1981) and a residual primary curvature of about 70°, inherited from the Precambrian basement. Any palaeomagnetic result from the Iberian Meseta predating the tectonic closure of the arc must therefore be corrected for the proper amount of rotation undergone during the Hercynian deformation. In addition, an independent rotation has occurred during the Mesozoic opening of the Bay of Biscay and palaeomagnetic directions must also be corrected for that 35° rotation (Van der Voo 1969) before any comparison with 'stable Europe' data can be made.

The Asturian arc forms the inner part of the Ibero–Armorican arc. Structural studies (Julivert 1971; Bonhommet *et al.* 1981) have shown that folding has occurred in two phases. The older (Middle Carboniferous) cylindrical folds, tangential to the arc, have been reshaped by later, radial, folds associated with the tectonic closure of the arc, resulting in interference patterns. The eight sites (56 samples) studied here are distributed all around the cartographic nose of a plunging fold (Fig. 1), in order to investigate possible unfolding procedures. The San Pedro Formation consists of quartzitic beds with a high haematite content (10–67 per cent). Beds are composed of detrital well-rounded quartz grains often coated as haematitic ooids. The original cement (quartz) is often replaced by haematite, which means that this haematite post-dated the deposition of the rock. No evidence has been found for strain inside the beds.

Palaeomagnetic measurements

Natural remanent (NRM) measurements have been performed with a computer-assisted Schonstedt spinner magnetometer. Alternating field (AF) and thermal demagnetizations have been done with Schonstedt equipment. Both these treatments were performed in field-free space with residual fields lower than 10 nT. Chemical demagnetization has been executed following the Park's (1970) procedure. No field-free space was available for storing samples during leaching. To check viscous effects, sample orientations were inverted from one step to another. This method lead to oscillations in the vertical projection of the Zijderveld demagnetization diagrams (in sample coordinates), but avoids any systematic

deviation of viscous origin. After a laboratory viscosity test (Thellier & Thellier 1959), showing very low short-period viscosity coefficients (< 5 per cent), all samples have been subjected to progresive thermal, chemical and AF demagnetizations. The results have been displayed and graphically interpreted on Zijderveld diagrams, and analysed by the great circle method (Halls 1978) when necessary. Where two procedures have been performed on the same sample (on duplicate specimens), simultaneous interpretation of the data has given one determination for a component of magnetization within a sample. As AF demagnetization had little effect on the NRM, measurements have been concentrated on thermal (46 specimens) and chemical procedures (30 specimens).

Results

The following behaviour has been observed, varying from site to site:

(1) In sites G, H, I and K, only one component of magnetization appeared, with a regular progression of the vector end-points towards the origin in Zijderveld plots, for chemical as well as for thermal procedures. Characteristic directions are very well clustered within each site ($k > 100$).

(2) In sites L, M, N the NRM is clearly of a bivectorial nature. Two components of magnetization can be obtained from linear segments (graphical determination) in the Zijderveld plots (Fig. 2). The thermal and chemical spectra do not seem to overlap entirely since the curves reveal angular bends. The direction of the high-temperature component (> 650°C) is the same as the direction first removed during leaching (< 500 hr) (Fig. 2) and is associated with the pigment which disappears completely during leaching. This is the only one self-consistent component within each site ($k > 40$) and is thought to be the charateristic one.

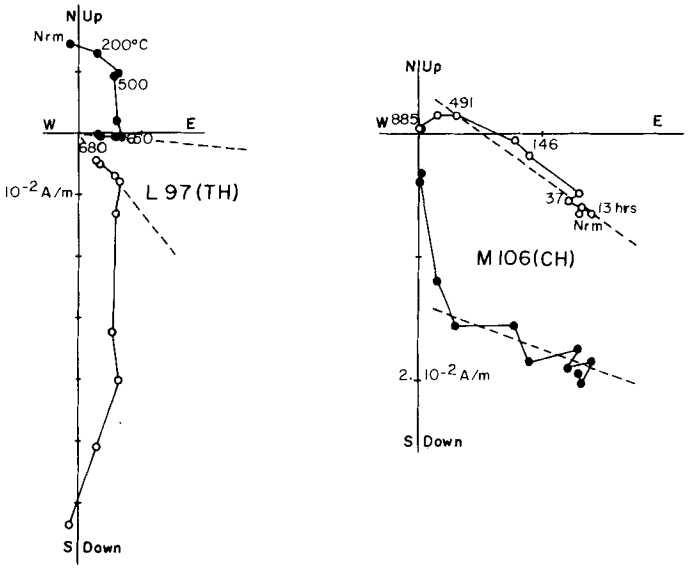


Figure 2. Zijderveld vector end-point diagrams for a thermal (left) and a chemical (right) demagnetization from sites where two components are present. The full (open) dots are for the projection into the horizontal (vertical) plane. Data are shown after tilt correction. The dashed lines represent the characteristic component.

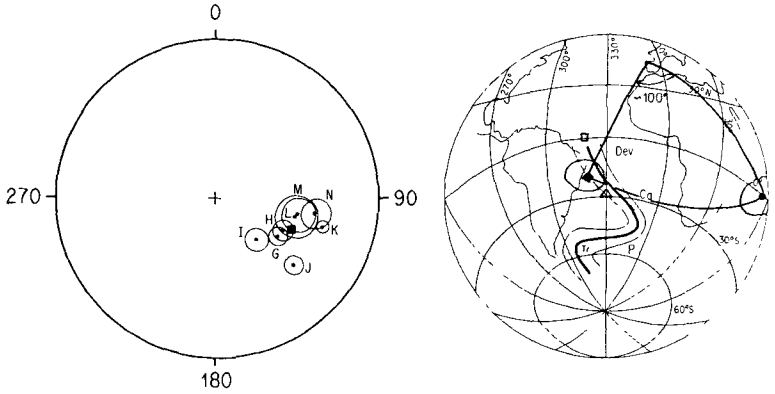


Figure 3. Left: stereographic projection of the site-mean directions after cleaning and tilt correction. ■ represents the formation mean direction. Right: comparison of the European polar wander path with the Cabrillanes result (● before, ♦ after geodynamic rotations). □ and △ represent the two areas of concentration of the European Lower Devonian poles as described in the text.

(3) In site J, Zijderveld plots reveal only one linear trend but this generally does not go towards the origin. This indicates at least two components of magnetization which we are unable to separate completely. The 'apparent' component (the one which corresponds to the linear trend) could be either one of them, isolated by the treatments, or a mixture of both. Great circle analysis allows the determination of one magnetization plane for each sample, in the case of two non-(anti) parallel components (Halls 1978). These planes intersect fairly well (quality factor $Q = 7 \times 10^{-6}$); the direction of intersection is statistically identical to the 'apparent components' Fisherian mean direction. We will therefore use that last direction with its circle of confidence as the mean characteristic direction for this site. As the planes diverge, the other component must be random in direction and will not be discussed further.

The eight site-mean directions (Fig. 3) with their statistical parameters are given in Table 1. Considered together, they show a positive fold test (k increases from 9 to 30 upon application of the tilt correction), significant at the 95 per cent confidence level. The mean direction is $D = 113°$, $I = +34°$, $\alpha_{95} = 10°$. In spite of the complete and detailed procedures and the good consistency of the results, the possibility remains that the characteristic magnetization are not completely cleaned. As a matter of fact, in some sites, the within-site characteristic direction distribution presents a 'streaking' that could correspond either to a

Table 1. Site-mean results of the palaeomagnetic analysis.

	Before/after tilt correction							Strike	Dip
Site	N	D	I	D	I	k	α_{95}		
G	7	120	−31	121	+40	122	05	202	72 NW
H	7	132	−09	115	+39	102	06	265	71 N
I	8	147	−32	134	+49	60	07	272	83 S
J	7	132	−33	130	+24	148	05	203	60 NW
K	6	107	−28	105	+20	489	03	165	55 W
L	6	102	−34	103	+35	41	12	141	84 NE
M	7	77	−02	101	+34	41	10	113	77 S
N	8	113	+15	99	+25	62	07	272	38 N
Mean	8	116	−21			9	–		
Mean	8			113	+34	30	10		

small dispersion in a plane of simply to an insufficient number of data points. However, best-fitting planes for these sites converge (and only after tilt correction) towards a direction identical to the formation-mean direction. Thus, even if a small dispersion may be attributed to the presence of an unresolved parasitic magnetization, we are confident that our estimate of the formation-mean direction is valid.

Discussion

As mentioned earlier, the characteristic magnetization is probably carried by haematitic pigment since the colouration disappears together with the magnetization during the chemical leaching. A chemical post-depositional origin is therefore inferred. On the other hand, the site-mean directions pass a fold test at the 95 per cent confidence level. Various unfolding models, taking into account a plunging fold axis, have been reported by Perroud & Cobbold (1984) but none of them led to definitive improvement of data grouping. A synstectonic origin has also been tested by using different amounts of partial unfolding but the k values for the eight site-mean directions never reached the 100 per cent tilt corrected level of 30. The age of the magnetization can therefore be assigned with confidence to the Lower Devonian–Middle Carboniferous time span. Early Carboniferous results from San Emiliano (Bonhommet *et al.* 1981), which is situated a few tens of kilometres from Cabrillanes, give a significantly shallower inclination ($I = 13° \pm 2$). Other early to Middle Carboniferous data for the Armorica plate (see Perroud & Bonhommet 1983, for a compilation) always reveal inclination lower than 20°. We can therefore assume that the magnetization in the San Pedro Formation has been acquired earlier than this period and must be of Devonian age. In the present state of knowledge about the development of haematitic pigment in redbeds (see Elmore & Van der Voo 1982 for a discussion) we are not able to restrict this interval further. However, no special event within the Devonian could be associated with haematite precipitation, and we prefer an acquisition time shortly after consolidation. We therefore consider the whole Devonian as a possible period for remanence acquisition, but we give the highest probability to an early Devonian age of the characteristic magnetization.

Although simple tilt correction on beds from a plunging fold (the plunge reaches 65° in some sites) could introduce declination errors, we estimate that the distribution of our sampling sites all around the nose of the fold leads to a rough cancellation of errors. This is probably one of the causes of the dispersion of the site-mean directions, as noticed by Perroud & Cobbold (1984), but is taken into account in the determination of α_{95} (= 10°).

Eastwards directions such as these reported here ($D = 113°$) are not uncommon in this region and have led to the recognition of the partially tectonic origin of the Ibero–Armorican arc. To compare our result with other European data, we first need to correct for this effect and also for the opening of the Bay of Biscay, which has resulted in a decrease of the declination by 35° (Van der Voo 1969). The former correction is done using the preliminary model proposed by Perroud & Bonhommet (1981) for the pre-orogenic shape of the Ibero–Armorican arc. Thus we obtain a corrected direction of magnetization $D = 209°$, $I = +34°$ for a locality situated at 44°N and 349°E. This corresponds to a palaeomagnetic pole at 22°S and 319°E ($dp = 6°$, $dm = 11°$). This pole is in fairly good agreement with available European poles for the same period (Fig. 3). In fact, it falls between a group consisting mostly of Siluro–Devonian poles for Great Britain (clustered about a mean pole at 0°N, 320°E) and a group consisting of poles from other parts of stable Europe (East Germany, Checkoslovakia, Scandinavia and the Russian Platform), concentrated about a mean pole at 30°S, 330°E. On the other hand, it does not fit in with the proposed apparent polar wander paths for Gondwana, whatever the path (McElhinny & Embleton 1974 or Morel & Irving 1978).

Conclusions

The magnetization revealed in the red beds of the San Pedro Formation represents a (probably early) Devonian palaeofield and has recorded geodynamic rotations during the Hercynian orogeny (tectonic closure of the Ibero–Armorican arc) and during the opening of the Bay of Biscay. The palaeomagnetic pole calculated after the necessary corrections for these rotations shows a close affinity with other (more northerly) European data. We therefore conclude that the Armorica microplate, which was part of Gondwana in the Early Palaeozoic (Van der Voo 1982), had joined the northern continents (Laurentia–Baltica assembly) sometime before or during the Devonian, and has since moved as part of Europe. The palaeolatitude of 18°S, corresponding to the inclination of +34° in a reverse field ($D = 209°$), is also similar to Devonian results from Nova Scotia and the Acadian displaced terraines of the east coast of the North American continent (Kent & Opdyke 1978; Johnson & Van der Voo 1983). This suggests that by Devonian times, Europe, North America and mobile belts of Armorica were closely assembled in what will become Pangea after the collision with Gondwana. The large transcurrent motion proposed to have occurred between the American craton and the Acadia displaced terraines has therefore involved Armorica (and Baltica) as well (Van der Voo 1983).

The result obtained here is quite consistent with the pre-orogenic shape of the Ibero–Armorican arc proposed by Perroud & Bonhommet (1981). The secondary (tectonic) part of the present-day curvature is therefore believed to be closer to 80°, as proposed by us, than to 110° as proposed by Ries *et al.* (1980).

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References

Bonhommet, N., Cobbold, P. R., Perroud, H. & Richardson, A., 1981. Paleomagnetism and cross-folding in a key area of the Asturian arc (Spain), *J. geophys. Res.*, **86**, 1873–1887.

Cogné, J., 1971. Le Massif Armoricain et sa place dans la structure des socles Ouest-Européens: l'arc Hercynien Ibero-Armoricain, in *L'histoire structurale du Golfe de Gascogne*, tome I, IFP édition, II–I–II–23.

Elmore, D. J. & Van der Voo, R., 1982. Origin of hematite and its associated remanence in the Copper Harbor Conglomerate (Keweenawan), Upper Michigan, *J. geophys. Res.*, **87**, 10 918–10 928.

Halls, H. C., 1978. The use of converging remagnetization circles in palaeomagnetism, *Phys. Earth planet. Int.*, **16**, 1–11.

Johnson, R. & Van der Voo, R., 1983. Paleomagnetism of the Mid–Devonian Fisset Brook Formation, Cape Breton island, Nova Scotia, *Eos, Trans. Am. geophys. Un.*, **64**, 216.

Julivert, M., 1971. L'évolution structurale de l'arc Asturien, in *L'histoire Structurale du Golfe de Gascogne*, tome 1, IFP édition, 12–1–12–28.

Kent, D. V. & Opdyke, N. D., 1978. Paleomagnetism of the Devonian Catskill redbeds: evidence of the motion of the coastal New-England–Canadian Maritim region relative to cratonic North America, *J. geophys. Res.*, **83**, 4441–4450.

McElhinny, M. W. & Embleton, B. J. J., 1974. Australian palaeomagnetism and the Phanerozoic plate tectonics of Eastern Gondwana, *Tectonophys.*, **22**, 1–29.

Matte, P., 1974, Raccord des segments hercyniens d'Europe Sud-Occidentale. Franz Kossmatt Symposium, *Nova Acta Leopoldina*, **224**, 239–262.

Morel, P. & Irving, E., 1978. Tentative palaeocontinental maps for the Early Phanerozoic and Proterozoic, *J. Geol.*, **86**, 535–561.

Paris, F. & Robardet, M., 1977. Paleogeographie et relations Ibero-Armoricaines au Paleozoique anté-Carbonifère, *Bull. Soc. géol. Fr.*, **7**, 19, 5, 1121–1126.

Park, J. K., 1970. Acid leaching of redbeds and the relative stability of the red and black magnetic components, *Can. J. Earth Sci.*, **7**, 1088–1092.

Perroud, H. & Bonhommet, N., 1981. Palaeomagnetic of the Ibero-Armorican arc and the hercynian orogeny in Western Europe, *Nature*, **292**, 445–448.

Perroud, H. & Bonhommet, N., 1983. Paleomagnetic evidence for lithospheric deformation in the European Hercynian foldbelt, 18th I.U.G.G., Hambourg prog. and abst., vol. 1 : 104.

Perroud, H., Bomhommet, N. & Van der Voo, R., 1983. Geographic extent and paleozoic evolution of the Armorica plate on the basis of paleomagnetic data, *I.A.G.A. Bull.*, **48**, 172.

Perroud, H. & Cobbold, P. R., 1984. L'aimantation rémanente comme marquer de la déformation: example d'un pli à axe incliné dans les séries rouges siluro-dévoniennes à Cabrillanes, Asturies, Espagne, *Bull. Soc. géol. Fr.*, submitted.

Ries, A. C., Richardson, A. & Shackleton, R. M., 1980. Rotation of the Iberian arc: paleomagnetic results from North Spain, *Earth planet. Sci. Lett.*, **50**, 125–135.

Thellier, E. & Thellier, O., 1959. Sur l'intesité du champ magnétique terrestre dans le passé historique et géologique, *Annls Géophys.*, **15**, 285–376.

Van der Voo, R., 1969. Palaeomagnetic evidence for the rotation of the Iberian peninsula, *Tectonophys.*, **7**, 5–56.

Van der Voo, R., 1982. Pre-Mesozoic paleomagnetism and plate tectonics, *Ann. Rev. Earth planet. Sci.*, **10**, 191–200.

Van der Voo, R., 1983. Paleomagnetic constraints for the assembly of the Old Red Continent, *Tectonophys.*, **91**, 271–283.